A Tale of Two Grammars: A Cophonological Analysis of Iconic Phonology

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Nutshell

Sound vs Lexical Depictives?
How do we know?

What Grammar Governs it?

WARNING: Generative Contents

Sound vs Lexical Depictives?

1. Setting the Scenes
2. Introducing the Diagnostics Framework
3. Walking you through with the Data

1. Setting the Scenes

Ideophones?

Marked words that depict and mimic sensory imagery and vividness (Doke, 1935).

Linkage between abstract iconicity and the physical sound shape.

1. Setting the Scenes

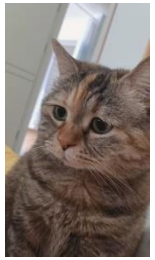
Ideophones?

Linkage between abstract iconicity and the physical sound shape.

Sound Depictives

Onomatopoeia

English: *meow*
Japanese: *nyaa*
Greek: *niáou*



Abstract Iconicity

German: *zack*

English: *bling*

1. Setting the Scenes

Ideophones?

Linkage between abstract iconicity and the physical sound shape.

(Source of vividness)

Sound Depictives + Reduplication

Japanese: *dokidoki / *doki*

"sound of heartbeat" AND "vividness"

1. Setting the Scenes

Cantonese Reduplication

Reduplication is a very common device for ideophonic derivations in Cantonese.

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Cantonese Reduplication

Reduplication is a very common device for ideophonic derivations in Cantonese.

Animal onomatopoeia (AA-*giu3*):

Cat: *meu1-meu1-giu3*, meow-meow-call
Dog: *wou-wou1-giu3*, bark-bark-call

1. Setting the Scenes

Cantonese Reduplication

Reduplication is a very common device for ideophonic derivations in Cantonese.

Abstract mimics (AA-*seng1*):

Hissing: *syut4-syut2-seng1*, hiss-hiss-sound
Gasping: *ham4-ham2-seng*, gasp-gasp-sound

1. Setting the Scenes

Cantonese Reduplication

Reduplication is a very common device for ideophonic derivations in Cantonese.

Nominal compounds with abstract mimics:

Pager: *bi1-bi1-gei1*, *bi1-bi1*-device

A type of tradition candy:

ding1-ding1-tong2,

ding1-ding1-candy

1. Setting the Scenes

Cantonese Reduplication

Reduplication is a very common device for ideophonic derivations in Cantonese.

Adjectival compounds come interesting:

Sparkling:
laap3-laap3-ling3, *laap3-laap3*-sparkling
Gloomy-cold:
jam1-jam1-dung3, gloom-gloom-cold

1. Setting the Scenes

Cantonese Reduplication

Sparkling:
laap3-laap3-ling3, *laap3-laap3*-sparkling
Gloomy-cold:
jam1-jam1-dung3, gloom-gloom-cold

They all share reduplication for vividness but...

1. Setting the Scenes

Cantonese Reduplication

This is clearly sound depictive

Sparkling:

laap3-laap3-ling3, ***laap3-laap3***-sparkling

Gloomy-cold:

jam1-jam1-dung3, gloom-gloom-cold

They all share reduplication for vividness but...

1. Setting the Scenes

Cantonese Reduplication

Sparkling:

laap3-laap3-ling3, *laap3-laap3*-sparkling

Gloomy-cold:

How about this?

jam1-jam1-dung3, **gloom-gloom**-cold

Is this sound or lexical depictive?

They all share reduplication for vividness but...

1. Setting the Scenes

Cantonese Reduplication

Sparkling:
laap3-laap3-ling3, laap3-laap3-sparkling
Gloomy-cold:
jam1-jam1-dung3, gloom-gloom-cold

In fact, apart of those depictive reduplication, Cantonese also employs prosaic reduplication.

Adjectival diminution (AA-*dei2*):

A little red:
hung4-hung5-dei2, red-red-DIM
A little scared:
geng1-geng1-dei2, scared.scared-DIM

1. Setting the Scenes

Cantonese Reduplication

Sparkling:
laap3-laap3-ling3, laap3-laap3-sparkling
Gloomy-cold:
jam1-jam1-dung3, gloom-gloom-cold

A little red:
hung4-hung5-dei2, red-red-DIM
A little scared:
geng1-geng1-dei2, scared.scared-DIM

1. Setting the Scenes

Cantonese Reduplication

Sparkling:
laap3-laap3-ling3, laap3-laap3-sparkling

Gloomy-cold:
jam1-jam1-dung3, gloom-gloom-cold

Sound depictive? Prosaics?

A little red:
hung4-hung5-dei2, red-red-DIM

A little scared:
geng1-geng1-dei2, scared.scared-DIM

1. Setting the Scenes

Research Questions

How to differentiate depictives and prosaics?

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My Proposal: with Syllable Fusion

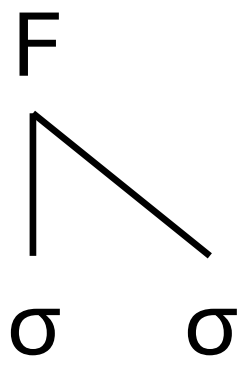
1. Setting the Scenes

Research Questions

How to differentiate depictives and prosaics?

Syllable Fusion

Two syllables blend together in rapid speech

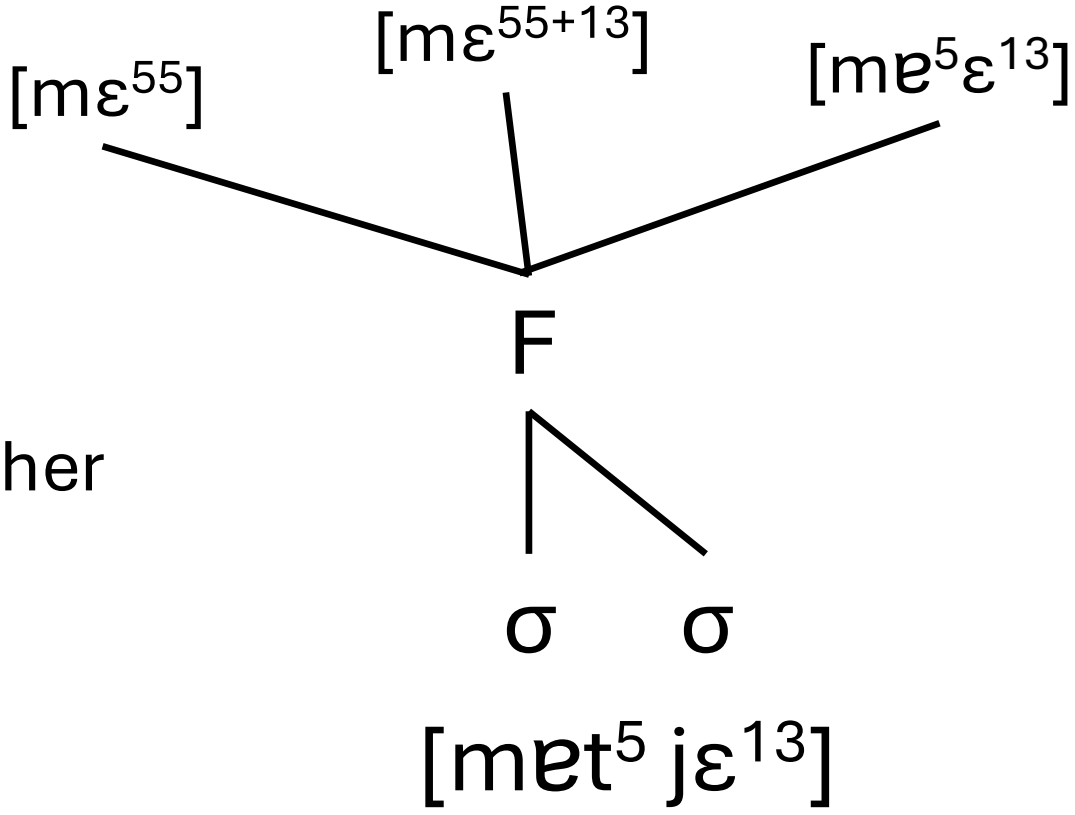


[mɐt^{5} jɛ^{13}]

1. Setting the Scenes

Research Questions

How to differentiate depictives and prosaics?



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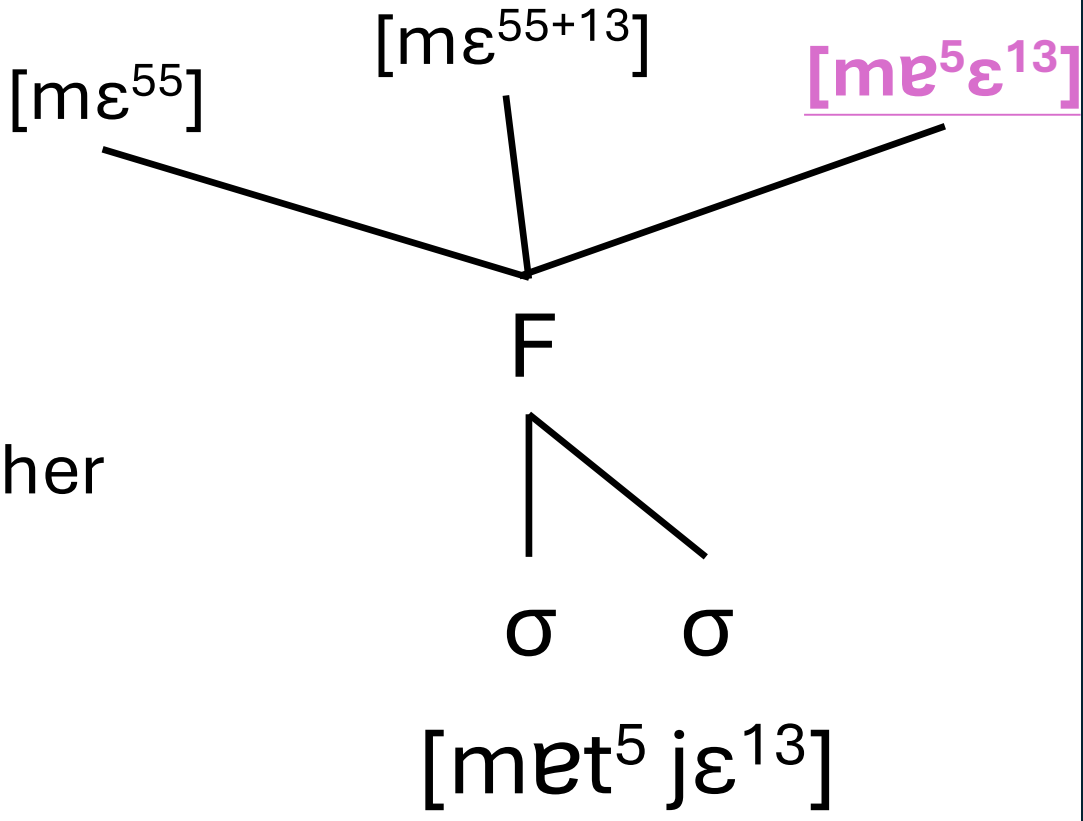

Syllable Fusion

Two syllables blend together in rapid speech

1. Setting the Scenes

Research Questions

How to differentiate depictives and prosaics?

Sound depictives

1. The physical sound shape is important for iconicity

2. Highly sensitive to their syllabic structure (Piggott, 2010; Kwon, 2017)

3. Syllable fusion destroys the syllabic integrity

X

4. Therefore, sound depictives would not fuse

Syllable Fusion

1. Setting the Scenes

Research Questions

How to differentiate depictives and prosaics?

Prosaics

1. Prosaics do not care much the physical sound shape

2. Therefore, prosaics should fuse just fine

Syllable Fusion

2. Introducing the Diagnostics Framework

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Some patterns in Cantonese syllable fusion

Syllable fusion is not arbitrary, it follows rules.

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Syllable fusion is not arbitrary, it follows rules.

The most important rule here: ALIGN-L(PWd, Ft)

2. Introducing the Diagnostics Framework

Some patterns in Cantonese syllable fusion

ALIGN-L(PWd, Ft)

nei4deon1-dou3 (Nathan-road)

2. Introducing the Diagnostics Framework

Some patterns in Cantonese syllable fusion

ALIGN-L(PWd, Ft)

nei4deon1-dou3 (Nathan-road)

Fusable: *neeon41*

2. Introducing the Diagnostics Framework

Some patterns in Cantonese syllable fusion

ALIGN-L(PWd, Ft)

nei4deon1-dou3 (Nathan-road)

Unfusable

The problem of Morphological Parsing?

2. Introducing the Diagnostics Framework

Some patterns in Cantonese syllable fusion

ALIGN-L(PWd, Ft)

zyu1-daai4coeng3 (pig-intestine)

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Unfusable

2. Introducing the Diagnostics Framework

Some patterns in Cantonese syllable fusion

ALIGN-L(PWd, Ft)

zyu1-daai4coeng3 (pig-intestine)

Fusable: *zyuaai14*

2. Introducing the Diagnostics Framework

Some patterns in Cantonese syllable fusion

ALIGN-L(PWd, Ft)

For any Prosodic Word (PWd),
only fuse the leftmost two syllables to form a Foot (Ft).

2. Introducing the Diagnostics Framework

Diagnostics Constraints

ALIGN-L(PWd, Ft) brings out a constraint

It can only test $[_{PWd}$ **XX**Y(Z)]

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ALIGN-L(PWd, Ft) brings out a constraint

It can only test $[_{PWd}$ XXY(Z)]

Not Testable:

Adjectival compounds: *dung3-bing1-bing1*

Quadrisyllabic mimics : *ding1-ding1-dong1-dong1*

2. Introducing the Diagnostics Framework

Diagnostics Constraints

ALIGN-L(PWd, Ft) brings out a constraint

It can only test $[_{PWd}$ XXY(Z)]

Testable:

Onomatopoeia: *wou1-wou1-giu3*

Abstract mimics: *ham4-ham2-seng1*

Nominal compounds: *ding1-ding1-tong2*

Adjectival compounds: *jam1-jam1-dung3*

Adjectival diminution: *geng1-geng1-dei2*

Quadrisyllabic mimics : *ding1-ding1-dong1-dong1*

2. Introducing the Diagnostics Framework

Diagnostics Items

Hypothesis:
Sound depictives cannot undergo Syllable Fusion as their syllabic shape is essential for necessary iconic depiction.

Onomatopoeia: *wou1-wou1-giu3*

Nominal compounds: *ding1-ding1-tong2*

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...

Test Set

2. Introducing the Diagnostics Framework

Diagnostics Items

Deideophonisation

Hypothesis:
Sound depictives cannot undergo Syllable Fusion as their syllabic shape is essential for necessary iconic depiction.

Test Set

Onomatopoeia: *wou1-wou1-giu3*

Nominal compounds: *ding1-ding1-tong2*

Adjectival compounds: *jam1-jam1-dung3*

Adjectival diminution: *geng1-geng1-dei2*

Quadrisyllabic mimics : *ding1-ding1-dong1-dong1*

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2. Introducing the Diagnostics Framework

Diagnostics Items

Deideophonisation

Hypothesis:
Sound depictives cannot undergo Syllable Fusion as their syllabic shape is essential for necessary iconic depiction.

Onomatopoeia: *wou1-wou1-giu3*

Nominal compounds: *ding1-ding1-tong2*

Verbalisation of sound depictives: *wou1-zo2* (V-PRF)

Adjectival diminution: *geng1-geng1-dei2*

Quadrisyllabic mimics : *ding1-ding1-dong1-dong1*

...

I wonder if it can fuse

2. Introducing the Diagnostics Framework

Diagnostics Items

Hypothesis:
Sound depictives cannot undergo Syllable Fusion as their syllabic shape is essential for necessary iconic depiction.

Onomatopoeia: *wou1-wou1-giu3*

Nominal compounds: *ding1-ding1-tong2*

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Adjectival diminution: *geng1-geng1-dei2*

Quadrisyllabic mimics : *ding1-ding1-dong1-dong1*

Verbalisation of sound depictives: *wou1-zo2* (V-PRF)

...

Test Set (Updated)

3. Walking you through with the Data

3. Walking you through with the Data

Experiment Setting

Acceptability Experiment

3. Walking you through with the Data

Acceptability Experiment

Reduplication Stimuli (Baseline)

Category	Sub-category	Items				
Sound depiction	Onomatopoeia	喵喵叫	咩咩叫	發發聲	雪雪聲	框框聲
	Nominal Compounds	啄啄糖	BB機	叮叮車	咔咔車	汪汪狗
	Quadrisyllabic Mimics	叮叮噹噹	兵兵乓乓	吱吱喳喳	鏗鏗哐哐	唏唏哈哈
Prosaics	Adjectival Diminution	怪怪哋	紅紅哋	臭臭哋	麻麻哋	甜甜地
	Adjectival Intensification	乾乾淨淨	奇奇怪怪	齊齊整整	開開心心	清清楚楚
	Verbal Discontinuation	食食下飯	跑跑下步	行行下街	聽聽下歌	傾傾下計

3. Walking you through with the Data

Acceptability Experiment

Reduplication Stimuli (Target)

Category	Sub-category	Items				
Adjectival Compound	Seemingly sound depiction	立立令	卜卜脆	騰騰震	瓜瓜嘈	戚戚痛
	Seemingly lexical	陰陰凍	濕濕碎	靜靜雞	赤赤痛	快快脆
De-depiction	Verbalisation	喵咗	咩咗	汪咗	發咗	砰咗

3. Walking you through with the Data

Procedure

26 native Cantonese speakers are recruited, ranging from 17-60 years old

3. Walking you through with the Data

Procedure

26 native Cantonese speakers are recruited, ranging from 17-60 years old

Unfused Sentence → Fused Sentence

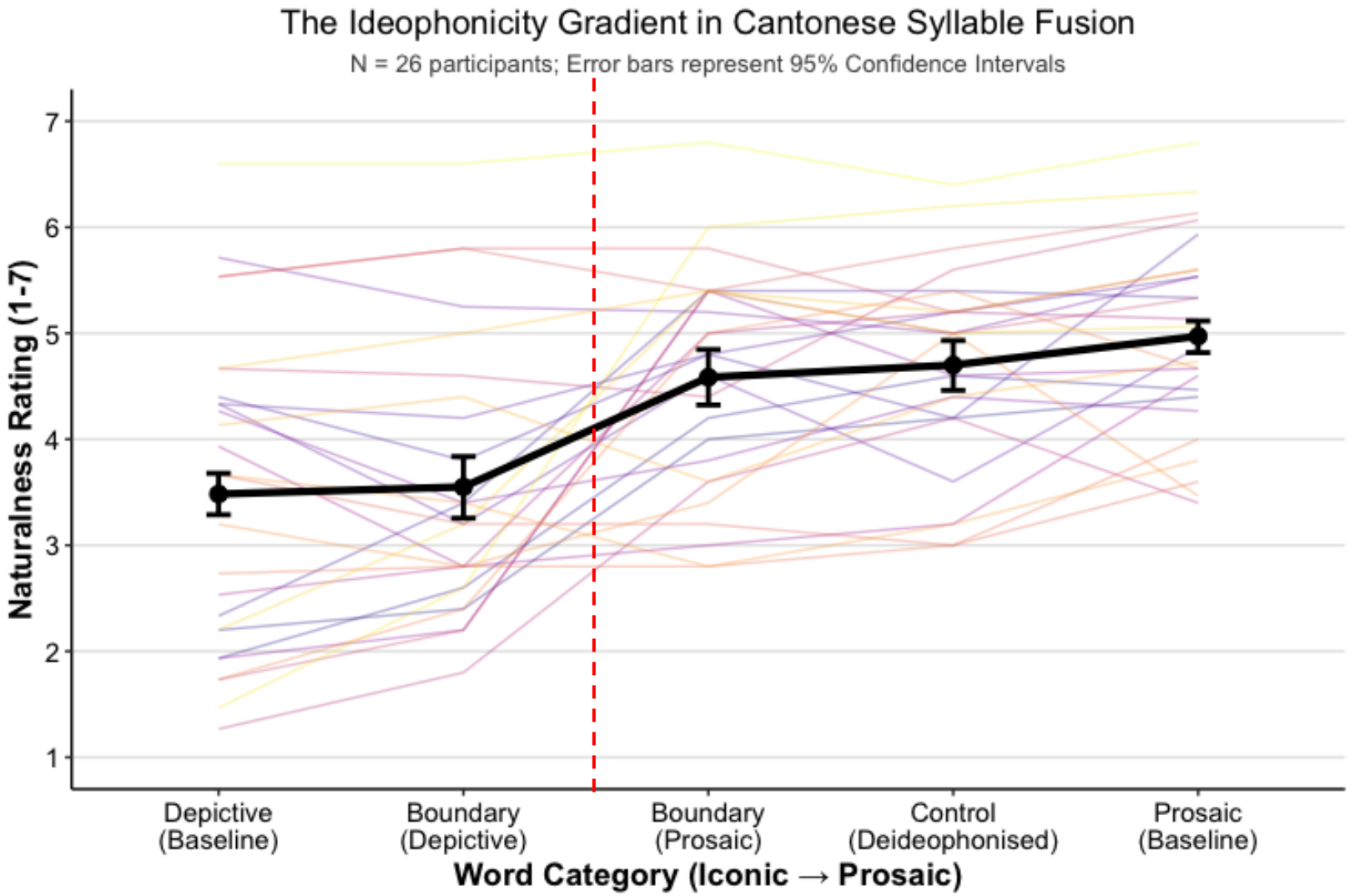
我聽到隻貓喺度 [meu1-meu1-giu3] 呀

我聽到隻貓喺度 [me:u1-giu3] 呀

Judge the acceptability of the fused sentence on 7-point Likert scale

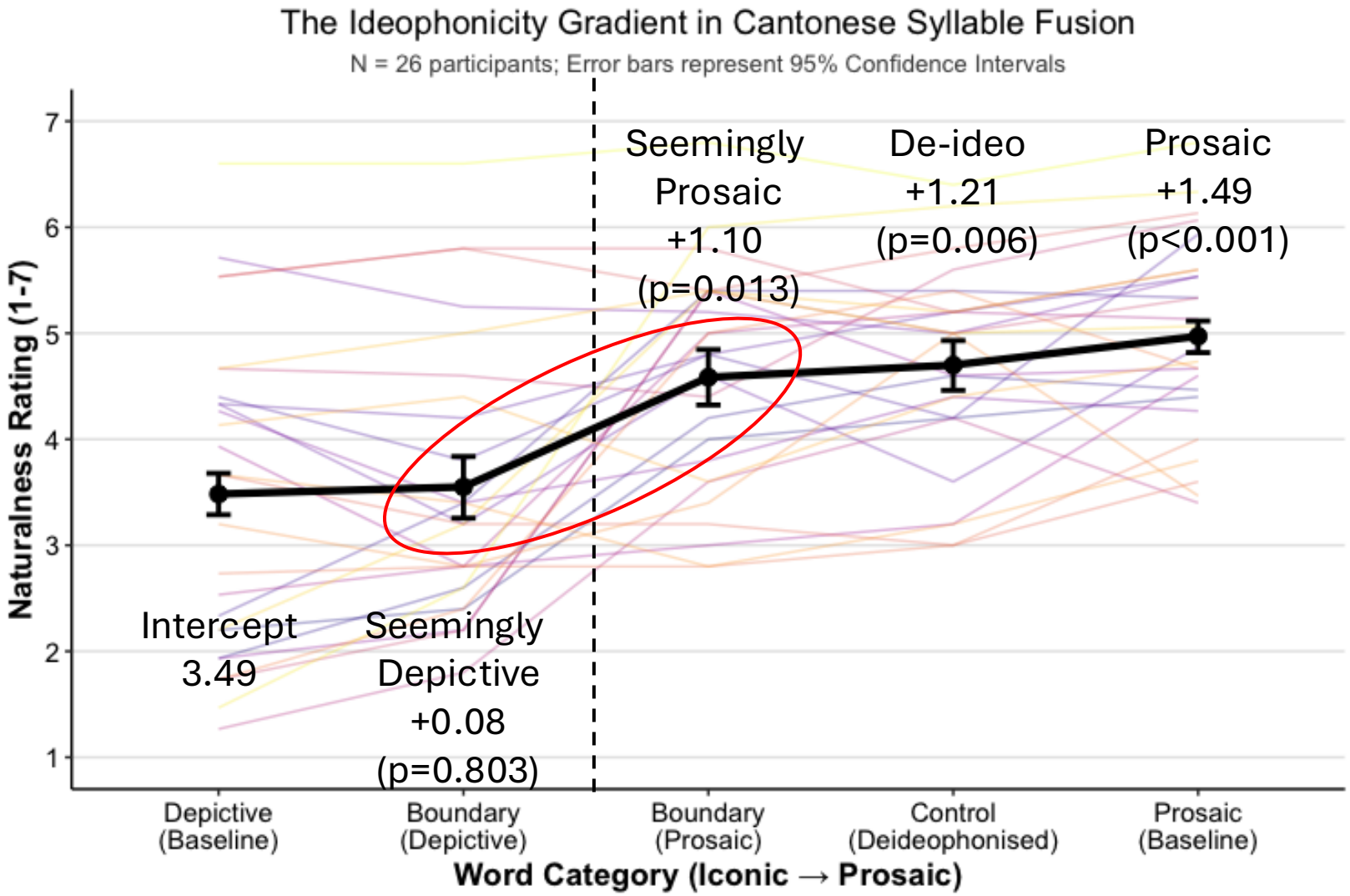
3. Walking you through with the Data

Result



3. Walking you through with the Data

Result



3. Walking you through with the Data

Result

Random Slope LMM (n=26)

Predictor	Estimate (β)	SE	*df*	*t*	*p*
(Intercept)	3.49	0.32	35.41	10.82	<0.001***
Seemingly depictive	0.08	0.31	45.29	0.25	.803 (n.s.)
Seemingly prosaic	1.10	0.43	50.37	2.57	.013*
De-ideophonised	1.21	0.42	50.66	2.88	.006**
Prosaic Baseline	1.49	0.33	46.91	4.57	< .001***

3. Walking you through with the Data

Result

Random Slope LMM (n=26)

Random Effect Group	Parameter	Variance
Item	(Intercept)	0.2744
Participants	(Intercept)	2.1380
	Seemingly depictive	**0.2212**
	Seemingly prosaic	2.4656
	De-ideophonised	2.3807
	Prosaic Baseline	1.6318

Depictives do not fuse under high consensus

3. Walking you through with the Data

Confounding Factors

Tone change

Frequency

Syllabic structure

Efficiency of Coarticulation

3. Walking you through with the Data

Confounding Factors

Tone change

Frequency

Nonetheless, a clear categorial correspondence can still be shown clearly.

Syllabic structure

Efficiency of Coarticulation

3. Walking you through with the Data

Conclusion

Sound depictives cannot fuse *laap3-laap3*-sparkling

Prosaics can fuse *gloom-gloom*-dung3

What Grammar Governs it?

1. OT and Cophonology
2. Distributed Morphology

1. OT and Cophonology

Optimality Theory

Sound forms are the outcome of ranking constraints

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Optimality Theory

Sound forms are the outcome of ranking constraints

ALIGN-L(PWd, Ft)

For any Prosodic Word (PWd),
only fuse the leftmost two syllables to form a Foot (Ft).

1. OT and Cophonology

Optimality Theory

Sound forms are the outcome of ranking constraints

ALIGN-L(PWd, Ft) » Many others

For any Prosodic Word (PWd),
only fuse the leftmost two syllables to form a Foot (Ft).

Very strong prosodic constraints
Once violated, it becomes unacceptable.

1. OT and Cophonology

Optimality Theory

Sound forms are the outcome of ranking constraints

ALIGN-L(PWd, Ft)

Markedness Constraint:

NO-HIATUS

If adjacent vowels are not across syllable boundaries, assign a violation.

A driving force to push grammar to fuse

1. OT and Cophonology

Optimality Theory

Sound forms are the outcome of ranking constraints

ALIGN-L(PWd, Ft) NO-HIATUS

Faithfulness Constraint:
UNIFORMITY (McCarthy & Prince, 1995)

If the output corresponds to more than one element in the input, assign a violation.

A resistant force to balance the grammar

1. OT and Cophonology

Optimality Theory

Syllable fusion in typical cases

ALIGN-L(PWd, Ft) / NO-HIATUS » UNIFORMITY

1. OT and Cophonology

Cophonology

Cophonology

ALIGN-L(PWd, Ft) / NO-HIATUS » UNIFORMITY

Some morphemes will trigger their own miniature OT grammar

1. OT and Cophonology

Cophonology

Default-Cophonology

ALIGN-L(PWd, Ft) / NO-HIATUS » UNIFORMITY

1. OT and Cophonology

Cophonology

Default-Cophonology

ALIGN-L(PWd, Ft) / NO-HIATUS » UNIFORMITY

Markedness » Faithfulness

1. OT and Cophonology

Cophonology

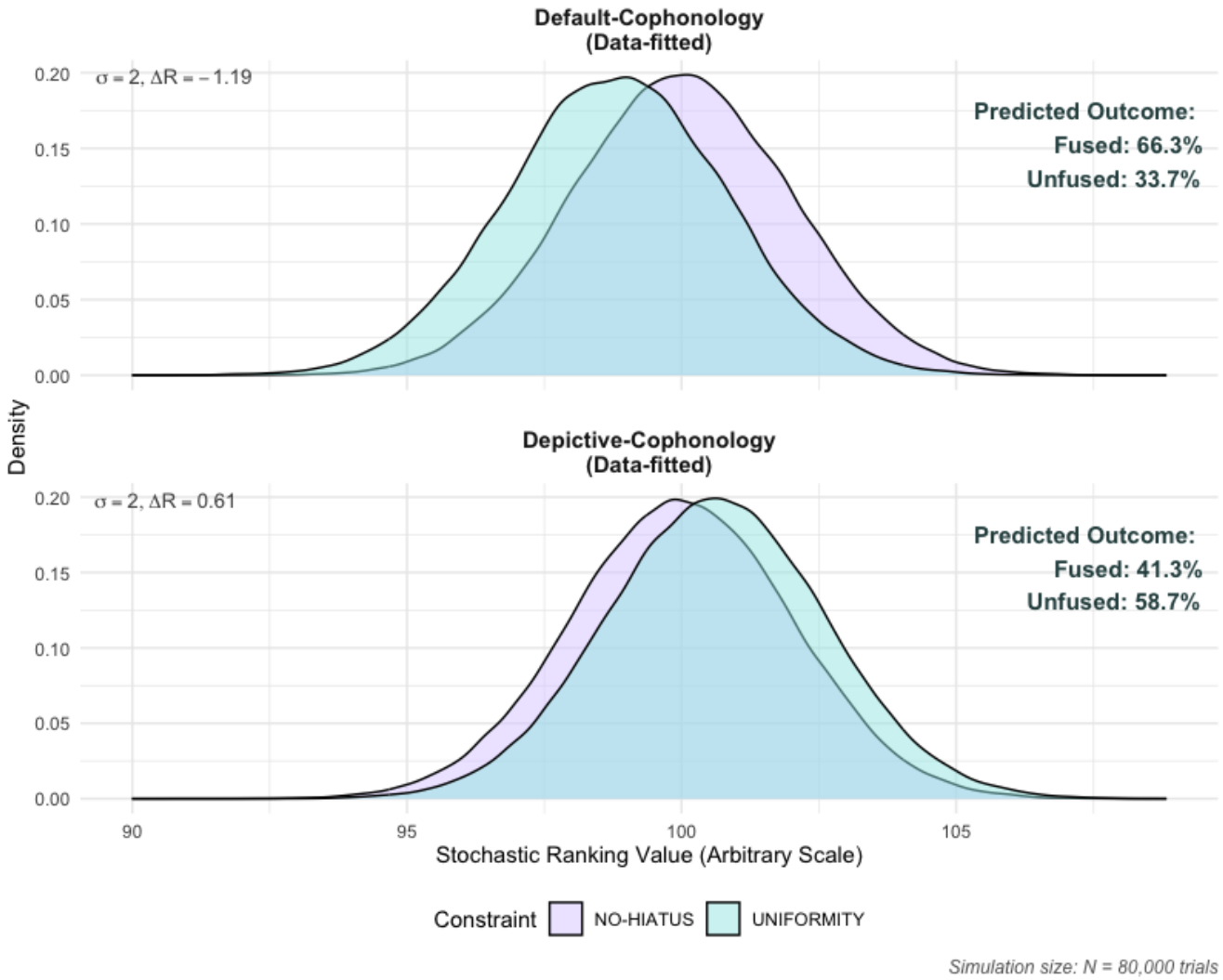
Sound Depictive-Cophonology

ALIGN-L(PWd, Ft) / UNIFORMITY » NO-HIATUS

Faithfulness » Markedness

1. OT and Cophonology

Stochastic Cophonology



2. Distributed Morphology

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The [DEPIC] Feature

The Cophonology for sound depictives requires a formal trigger

2. Distributed Morphology

The [DEPIC] Feature

The Cophonology for sound depictives requires a formal trigger

[DEPIC]

Just a label

2. Distributed Morphology

The [DEPIC] Feature

I propose a **Monosyllabic Root Hypothesis**

Depic^0 √MEOW

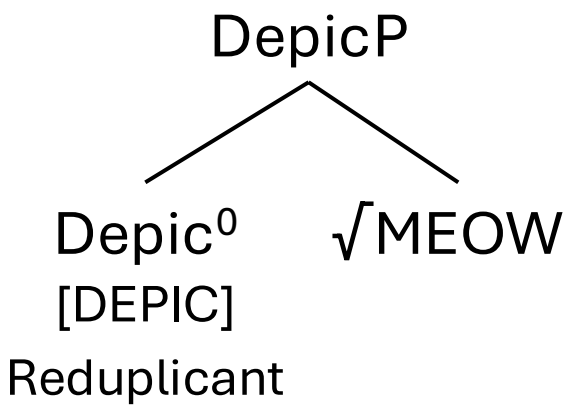
[DEPIC]

Reduplicant

2. Distributed Morphology

The [DEPIC] Feature

Merging a monosyllabic root



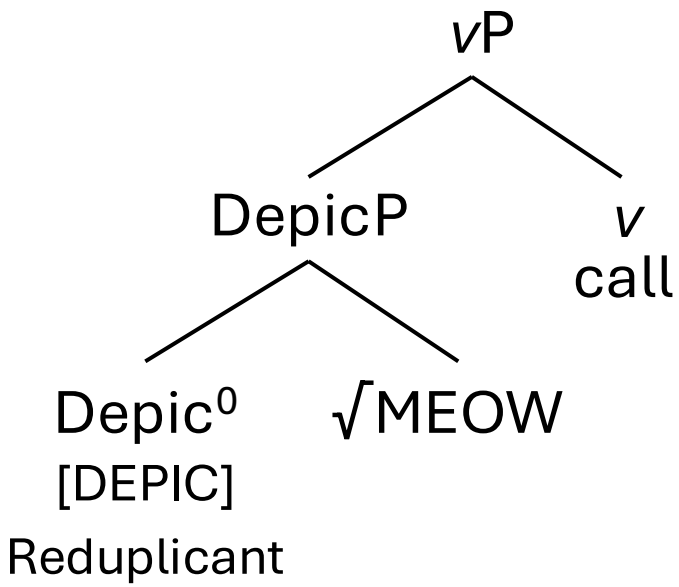
Sound depictives are always reduplicated

The Depic^0 head licenses the reduplicant morpheme

2. Distributed Morphology

The [DEPIC] Feature

Merging subsequent elements

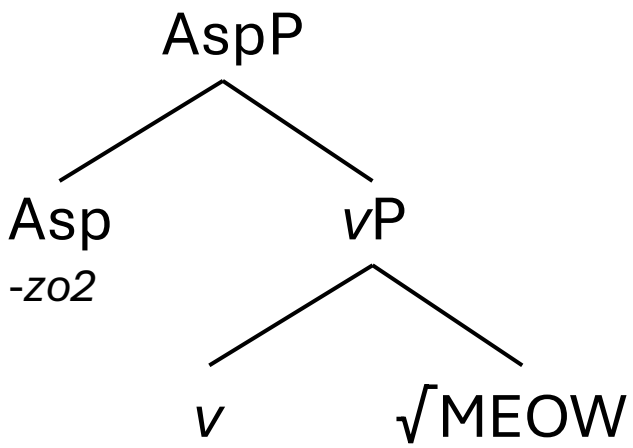


1. It ensures reduplication: *meu1-meu1-giu3* (*meow-meow-call)*
2. It ensures obedience to Depictive-Cophonology

2. Distributed Morphology

The [DEPIC] Feature

Under deideophonisation



1. It does not reduplicate anymore: *meu1-zo2* (*meow*-ASP$_{PRF}$)
2. It now accepts syllable fusion

Nutshell

Syllable fusion diagnoses that Cantonese sound depictives systematically ensures phonological integrity

Sound depictives trigger a micro-OT that ranks Faithfulness » Markedness for fulfilling the integrity

How about your language?

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